

Cells, Tissues & Body Systems

Your body is built from about 37 trillion cells — and every one of them is a tiny machine doing a job. Stack those cells together the right way and you get tissues, organs, and the systems that keep you alive. Let's zoom in.

 Georgia Standard S7L2

WHAT YOU'LL LEARN

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|------------------------------------|------------------------------------|
| 01 The Cell as a System | 05 Levels of Organization |
| 02 Animal Cell Organelles | 06 The Body's Systems |
| 03 Plant Cell Extras | 07 Systems Working Together |
| 04 How Organelles Cooperate | 08 Putting It All Together |

2.1 The Cell as a System

A **cell** is the smallest unit of life. But a cell isn't just a blob — it's a tiny **system** with many parts, each doing a specific job. Together, those parts let the cell get nutrients, grow, reproduce, build materials, and remove waste.

Think of a cell like a factory. The factory has a manager's office, a power plant, a packing department, and a cleanup crew. Each part is called an **organelle** — a "little organ" inside the cell. When every organelle does its job, the whole cell stays alive and healthy. This is what scientists mean when they say the cell works as a **system**.

! Did You Know?

If you lined up all the cells in your body end to end, they would stretch around the Earth more than 600 times. Yet a single cell is far too small to see without a microscope.

X Common Mix-Up

Cells and atoms are not the same thing! An **atom** is the smallest unit of *matter*. A **cell** is the smallest unit of *life* — and it's made of billions of atoms.

2.2 Animal Cell Organelles

Let's meet the main organelles found in an animal cell. Each one has a special job that keeps the cell running.

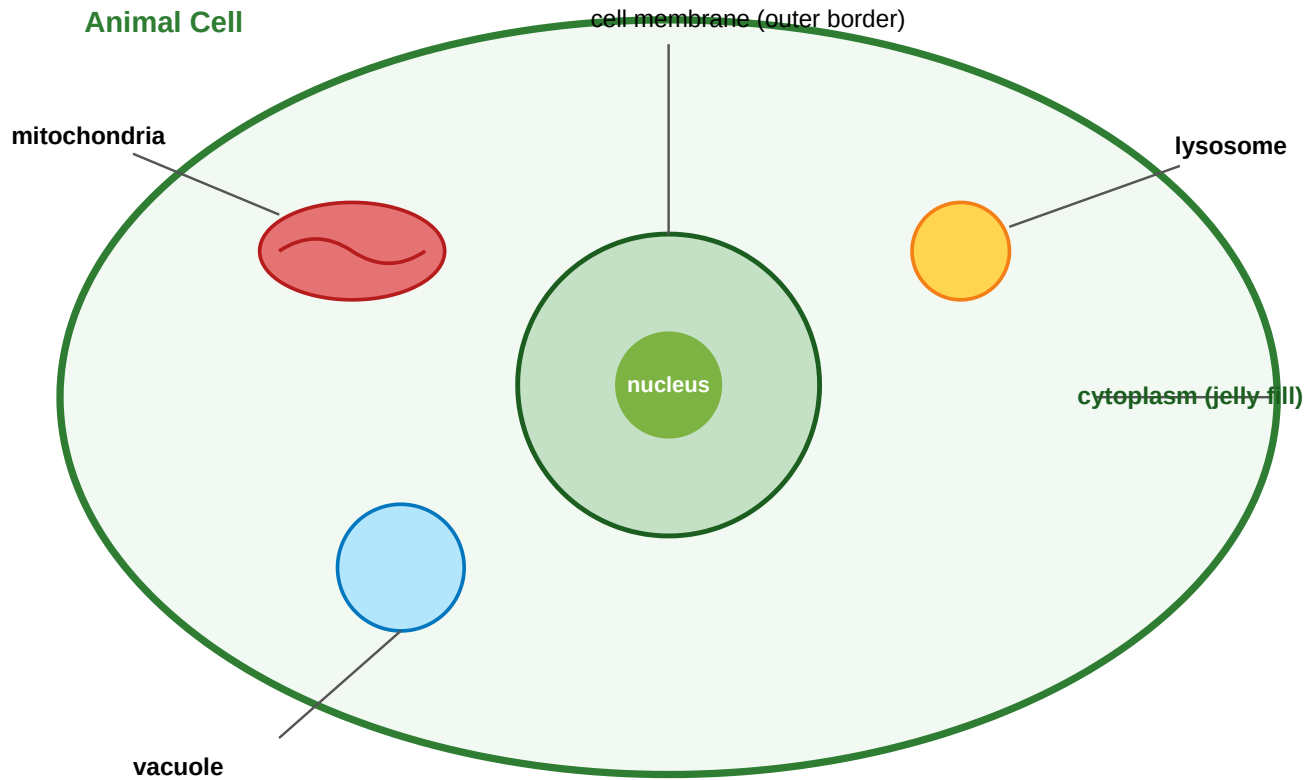


Figure 2.1 — A typical animal cell and its organelles.

 Nucleus Control center; holds DNA and gives instructions	 Cytoplasm Jelly that fills the cell; holds organelles in place	 Cell Membrane Outer border; controls what enters and leaves
 Mitochondria Powerhouse; releases energy from food	 Lysosome Cleanup crew; breaks down waste and worn-out parts	 Vacuole Storage; holds water, food, and waste

Key Idea — How the Cell Meets Its Needs

Each organelle helps the cell survive: the **membrane** lets in nutrients, the **mitochondria** turn that food into energy, the **nucleus** directs the cell to grow and reproduce, organelles build the materials the cell needs, and the **lysosome** processes waste.

2.3 Plant Cell Extras

Plant cells have everything animal cells have — plus three special parts that animal cells don't. These extras let plants stand tall and make their own food.

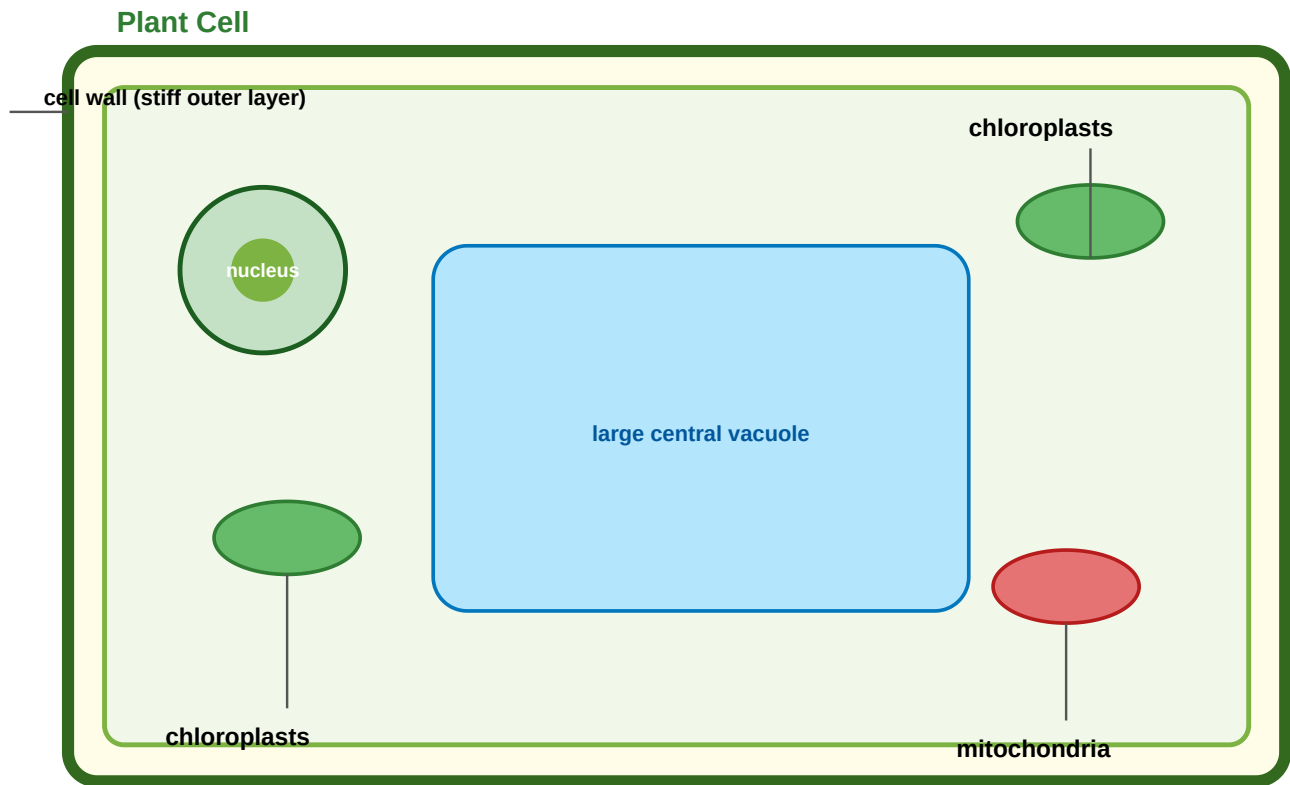


Figure 2.2 — A plant cell. Notice the cell wall, chloroplasts, and large vacuole.

Plant-Only Part	Job
Cell Wall	A stiff outer layer outside the membrane. Gives the plant cell its shape and support — it's why plants can stand up straight.
Chloroplasts	Green organelles that capture sunlight and make food through photosynthesis . They contain the green pigment chlorophyll.
Large Central Vacuole	A big storage sac filled with water. When it's full, it presses out and keeps the plant firm. When it's empty, the plant wilts.

★ Quick Compare

Both cells have: nucleus, cytoplasm, membrane, mitochondria, lysosome.

Only plants have: cell wall, chloroplasts, large central vacuole.

! Did You Know?

Chloroplasts are the reason plants are green and the reason you can breathe — photosynthesis releases the oxygen in our air.

2.4 How Organelles Cooperate

No organelle works alone. To do something as simple as grow, the cell needs many organelles working together as a system. Here's one example — making energy the cell can use:

★ **Teamwork Example: Getting Energy**

- The **cell membrane** lets a sugar molecule (food) enter the cell.
- The **cytoplasm** carries the sugar to where it's needed.
- The **mitochondria** break the sugar down and release energy.
- The **lysosome** cleans up the leftover waste.

One job — but four organelles cooperating. That's a system in action.

2.5 Levels of Organization

A single cell can't run an entire body on its own. Living things are built in levels: cells team up to form tissues, tissues form organs, organs form systems, and systems form a whole organism. Each level is more complex than the one before.

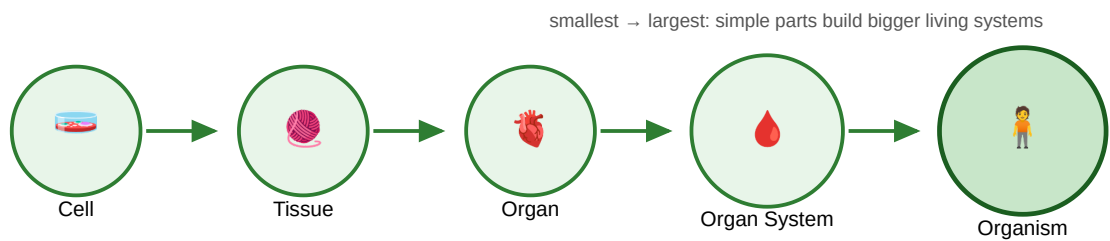


Figure 2.3 — The five levels of organization, from a single cell to a whole organism.

Level	What It Is	Example
Cell	The smallest unit of life	One muscle cell
Tissue	A group of similar cells doing the same job	Muscle tissue
Organ	Different tissues working together	The heart
Organ System	A group of organs with a shared task	Cardiovascular system
Organism	All the systems together — a complete living thing	A human

2.6 The Body's Systems

Your body runs on several major organ systems. Each has its own job, but none can work alone. Here are seven you should know:

System	Main Job	Key Organs
Cardiovascular	Pumps blood to carry oxygen and nutrients	Heart, blood vessels
Respiratory	Brings in oxygen, removes carbon dioxide	Lungs, trachea
Digestive	Breaks down food into nutrients	Stomach, intestines
Excretory	Removes liquid waste from the body	Kidneys, bladder
Muscular	Moves the body and its parts	Muscles
Nervous	Sends signals; controls and senses	Brain, nerves
Immune	Fights germs and disease	White blood cells, lymph nodes

! Did You Know?

Your nervous system sends signals at speeds up to 268 miles per hour — faster than a race car. That's how you can yank your hand off a hot stove before you even feel the pain.

2.7 Systems Working Together

The real magic happens when systems **interact**. To do any life process — like running across a field — several systems must cooperate at the same time.

★ Example: What Happens When You Run

- Your **respiratory** system pulls in extra oxygen.
- Your **cardiovascular** system pumps blood to carry that oxygen to your muscles.
- Your **muscular** system uses the oxygen and energy to move your legs.
- Your **nervous** system controls it all and keeps your balance.
- Your **excretory** system removes the waste your hard-working cells make.

Five systems, one action. If even one stopped, you couldn't run.

More Examples of Systems Interacting

These Systems...	Work Together To...
Digestive + Cardiovascular	Absorb nutrients from food, then deliver them to every cell through the blood.
Respiratory + Cardiovascular	Take in oxygen and carry it throughout the body.
Nervous + Muscular	Send the signal to move, then carry out the movement.
Immune + Cardiovascular	Move white blood cells through the body to fight infection.

2.8 Putting It All Together

You've zoomed all the way from a tiny organelle up to a whole living organism. Cells are systems made of cooperating organelles. Cells build tissues, organs, and systems. And those systems work together to keep an organism alive.

Big Idea	What to Remember
Cell as a system	Organelles cooperate so the cell can get nutrients, grow, reproduce, build materials, and process waste.
Animal organelles	Nucleus, cytoplasm, membrane, mitochondria, lysosome, vacuole.
Plant extras	Cell wall, chloroplasts, large central vacuole.
Levels	Cell → Tissue → Organ → Organ System → Organism.
Body systems	Cardiovascular, Respiratory, Digestive, Excretory, Muscular, Nervous, Immune.
Interaction	Systems must work together to carry out any life process.

★ Standard Check — S7L2

You can now **(a)** model how cell structures work as a system, **(b)** describe how cells are organized into tissues, organs, systems, and organisms, and **(c)** argue that body systems interact to carry out life processes. Whole standard — done!